Unequal Job Opportunities and the Measurement of Welfare Inequality

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France 2016, EU-SILC priced through EUROMOD.

Two questions.

(1) How much money-metric well-being inequality is attributable to unequal job opportunities rather than to preferences?

(2) When a model omits heterogeneous opportunities, how much of that inequality is re-attributed to preferences?

Three literatures, one gap.

Money-metric welfare with heterogeneous preferences

Fleurbaey and Maniquet (2005, 2011); Decoster and Haan (2015); Bargain, Decoster, Dolls, Neumann, Peichl and Siegloch (2013); Jacquet, Jia and Thoresen.

Random-utility random-opportunity labour supply

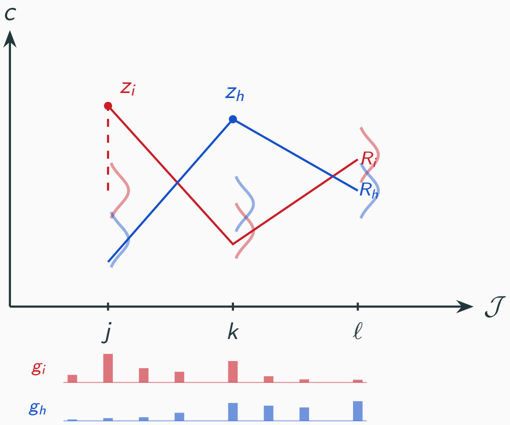
Dagsvik (1994); Aaberge, Colombino and Strøm (1999); Aaberge and Colombino (2013); Dagsvik and Jia (2016); Capéau et al.

Inequality of opportunity

Roemer (1998); Bourguignon, Ferreira and Menéndez (2007); Ferreira and Gignoux (2011).

This paper: household-specific latent job-opportunity distributions carried into a preference-respecting money-metric inequality decomposition, in both counterfactual directions.

Empirically the set is an estimated distribution and the pay is an offer density.



estimated jointly with preferences; nodes are integration support, not a list of jobs.

Preferences and an opportunity density, in one likelihood.

$$u_{ij} = \beta_\ell(x_i)\,\mathrm{BC}(l_{ij};\theta_\ell) + \mathrm{BC}(c_{ij};\theta_c)$$

$$g_{ij} = \underbrace{g^E}_{\text{whether work is available}} \cdot \underbrace{g^H}_{\text{at which hours}} \cdot \underbrace{g^{\mathrm{Occ}}}_{\text{in which occupation}} \cdot \underbrace{g^W}_{\text{at what pay}}$$

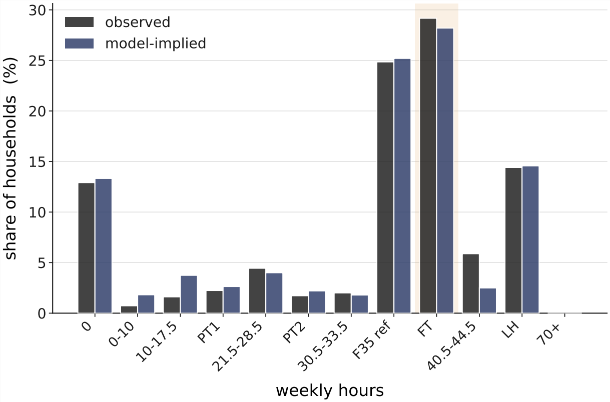
41 estimated structural parameters.

Each household: its observed job plus 100 drawn alternatives, all priced through the tax-benefit system.

$$V_{ij} = \underbrace{u_{ij}}_{\text{preferences}} + \underbrace{\log g_{ij}}_{\text{availability}} - \underbrace{\log q_{ij}}_{\text{sampling correction}}$$

EUROMOD FR 2016 $\rightarrow$ disposable income for every one of 157,055 alternatives.

The model reproduces the hours distribution, employment and occupation shares.



35-hour band 24.9% observed, 25.2% model-implied; employment 87.1% / 86.7%; mean absolute band error 0.008.

Equivalent income at a common reference pay, under four preference–environment states.

$$W_i^1 = w : \quad \Omega_i(\text{flat pay } w \text{ on own set}) = \Omega_i(\text{actual})$$

i.e. the uniform pay w over the household's own opportunity distribution that reproduces its attained expected welfare.

	own environment	reference environment
own preferences	I_{00}	I_{01}
reference preferences	I_{10}	I_{11}

reference environment = common job access, earning opportunities and budget; the same coalition's set and preferences on both sides of the inversion; common quadrature support.

A two-player Shapley game: preferences against the complete non-preference environment.

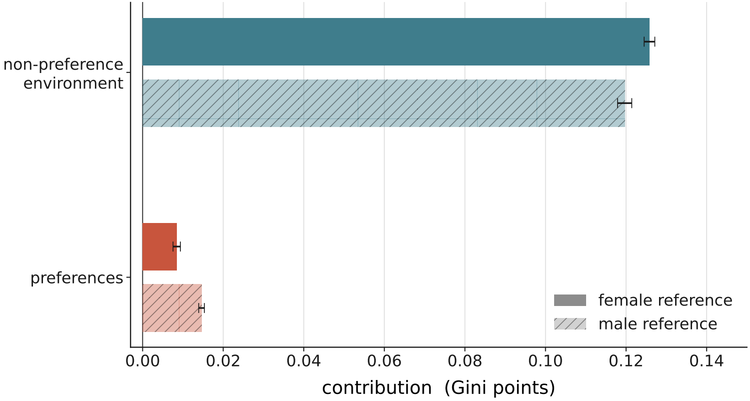
$$I_{00} = 0.134 \qquad I_{01} = 0.031$$
$$I_{10} = 0.148 \qquad I_{11} = 0.000$$

$$C_{\mathrm{pref}} = \tfrac{1}{2}[(I_{00} - I_{10}) + (I_{01} - I_{11})] = 0.0085$$
$$C_{\mathrm{env}} = \tfrac{1}{2}[(I_{00} - I_{01}) + (I_{10} - I_{11})] = 0.126$$

equalizing the environment alone reduces inequality by 77%, equalizing preferences alone raises it by 10%; the Shapley attribution averages the two orders and gives the environment 94%; composition: attributed 12.45%, equalized alone 2.00%.

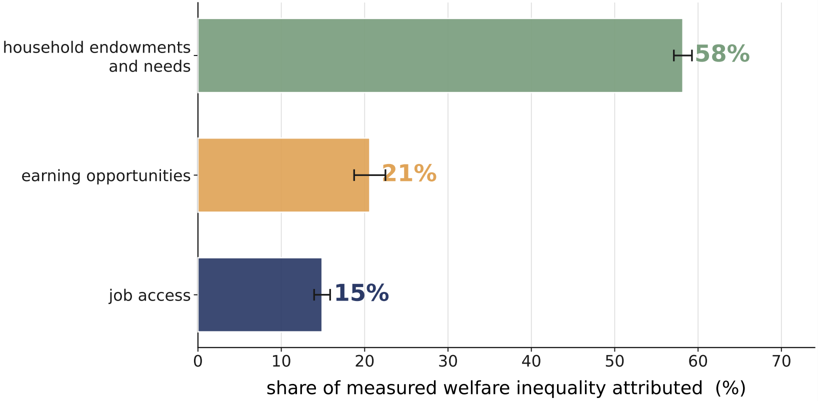
the environment is then split by an Owen value into job access, earning opportunities, and household endowments and needs.

The non-preference environment accounts for 94% of measured welfare inequality; preferences for 6%.



93.7 $\pm$ 0.6 / 6.3 $\pm$ 0.6 female-primary; 89.1 / 10.9 male structural-zero; never averaged. Parameter intervals: environment [89.1, 95.8], preferences [4.2, 10.9].

Household endowments and needs 58%, earning opportunities 21%, job access 15%; job opportunities together 36%.



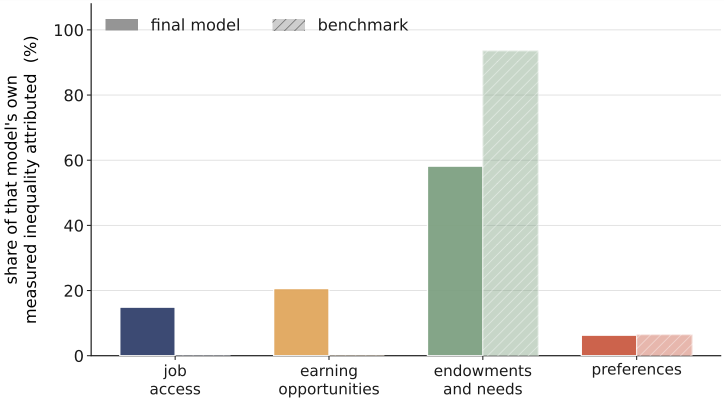
integration bands $\pm$ 1.5 / $\pm$ 1.6 / $\pm$ 1.1; ordering identical under both reference conventions and under equivalization.

Unequal job opportunities account for about a third of measured welfare inequality; omitting them re-attributes almost nothing to preferences.

(1) job access + earning opportunities: 35.5% $\pm$ 1.0 of measured inequality (the complete environment 93.7% $\pm$ 0.6).

(2) Omitting heterogeneous opportunities: preference share 6.3% $\rightarrow$ 6.4%; measured inequality falls 24.2%; the market-side contribution is relabelled into endowments and needs.

The benchmark fits the marginals as well, is 129 log-points worse, and recovers availability as taste.



35-hour coefficient: 2.58 as availability, 2.55 as preference; leisure-intercept sex gap $+0.43 \to -1.99$.

The preference share is the sensitive margin; the environment's internal structure is stable.

perturbation $\rightarrow$	preference share	environment ordering
reference convention	$6.3 \rightarrow 10.9$	unchanged
age-curvature bounds	$\approx +13\%$	unchanged
couples male-leisure pin	up to 73%	unchanged
50-400 drawn jobs	largest coefficient move 0.56 s.e. (0.2 s.e. from 100 to 400)	ordering unchanged

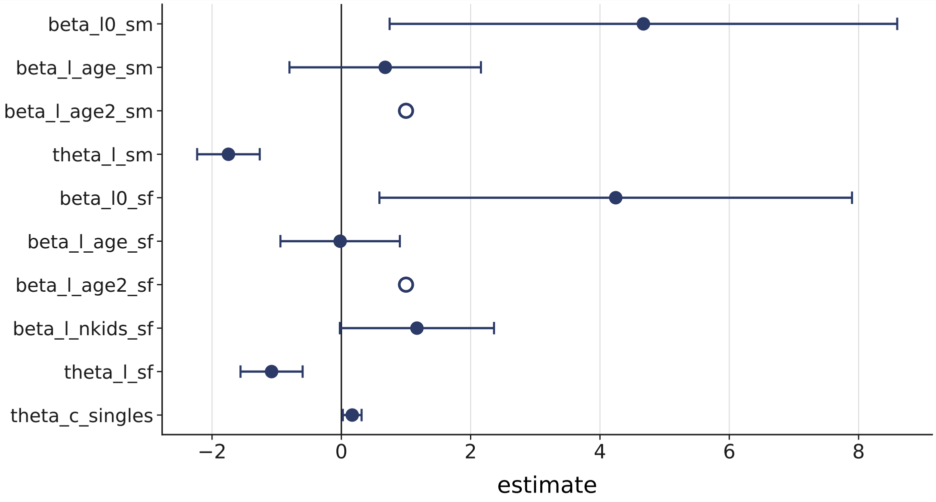
Conclusion

Method: a household-specific opportunity distribution estimated with preferences, priced through the real tax-benefit system.

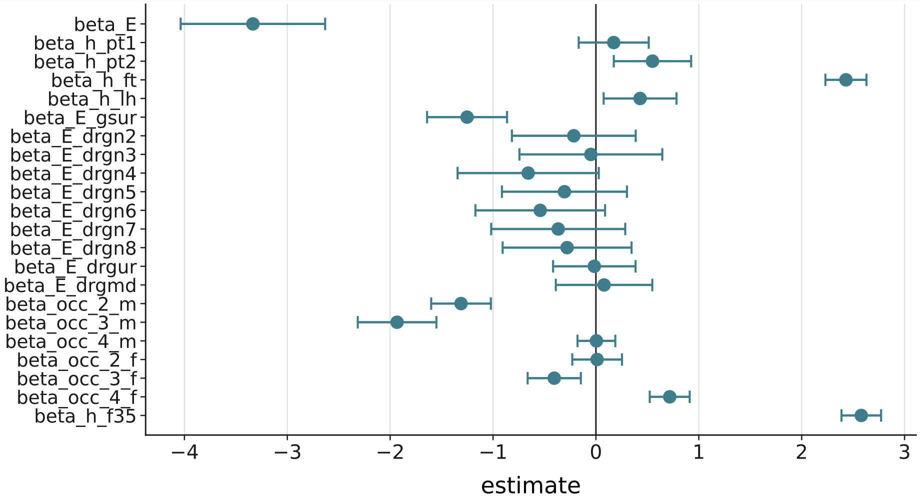
Finding: the non-preference environment carries 94% of measured money-metric inequality; job opportunities about a third; the budget side more than the market side.

Margin: the preference share is the sensitive quantity, 6 to 11 per cent across normative conventions.

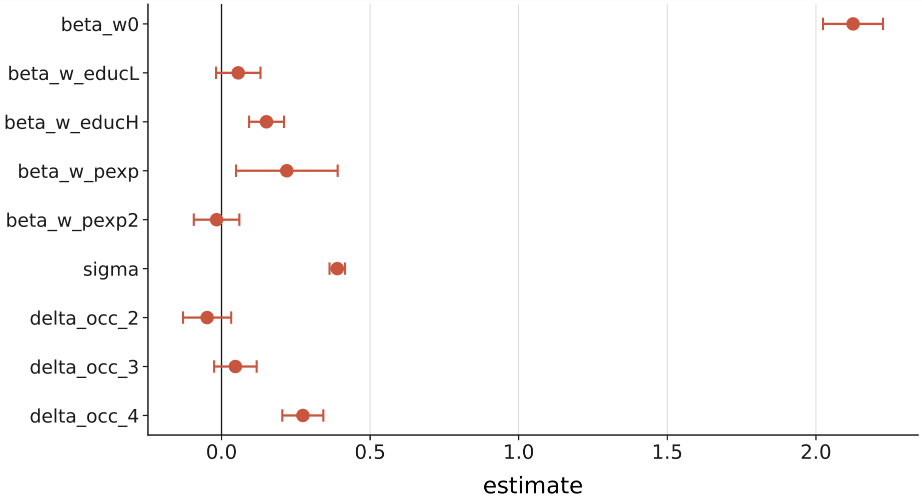
B1 — The estimated coefficients: preferences



B1 — The estimated coefficients: job access



B1 — The estimated coefficients: earning opportunities



B2 — Exhaustiveness is a tested property

preferences	job access
earning opportunities	household endowments and needs

The fully common state is 0.000000 — numerically zero.

B3 — The scale moves with the channel that equalises needs

	raw	equivalized
preferences	6.30%	7.98%
opportunities and budgets	93.70%	92.02%
job access	14.90%	9.68%
earning opportunities	20.62%	16.24%
household endowments and needs	58.18%	66.10%
market side	35.51%	25.92%

Freezing the own scale leaves exactly the Gini of its inverse.

B4 — the desired-hours counterpart test

ground	what it shows
definitional	the candidate's optimum tracks pay
identification	the hours block moves the moment by **exactly 0**
numerical	the level can agree; the profile cannot

Tried, and it fails on three independent grounds.

B5 — Where the welfare family comes from

Compensation

unequal productivity should not become unequal well-being

Responsibility

preference-based choices should not be compensated

$$\nexists\, W : \quad \text{Full Compensation} \;\wedge\; \text{Full Responsibility}$$

[Fleurbaey & Maniquet, *SCW* 2005]

W^1 is one compromise — the one this paper takes to the data.

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).

B6 — The RUM benchmark details

	final model	benchmark
35-hour coefficient	2.58	2.55
	as availability	as preference
leisure-intercept sex gap	+0.43	-1.99
preference share	6.3%	6.4%

Job access and earnings are exactly zero here. By construction.